

PARTNER BRIEFING · SEPTEMBER 2026

TACYOLO

Software-defined detection and tracking for cameras — and a radar path when you have one. Built to run on hardware you already own.

Seeking a company to productize, field, and own the first SKU.

Confidential. Dual-use computer vision. Not a targeting or weapons-release system.

01

Detect & classify

People, vehicles, drones, armor, aircraft

02

Track in time

EKF tracks with radar range/bearing when present

03

Ship on existing silicon

Windows prototype now · Jetson next · no new PCB

Cameras got cheap. Attention did not.

Small UAS, mixed traffic, and 24/7 towers produce more pixels than operators can watch. Off-the-shelf YOLO demos detect a class list. They do not decide when to spend GPU, keep identity, or fuse a second sensor.

The operator problem

Idle scenes still run full inference. Small movers are missed. Tracks break behind cover. A second sensor (radar, IR) is bolted on as a separate product.

The integrator problem

Defense and security OEMs do not want another GitHub notebook. They want a software stack that fits existing cameras, existing GPUs, and an existing C2 bus.

The honesty problem

The market is full of 99% claims, thermal magic, and “AI fire control.” Serious buyers discount that immediately. We designed this briefing the opposite way.

One stack. Software only. Existing hardware.

TACYOLO is a Python product wrapper around a gated detector, a gallery matcher, an EKF tracker, and a radar-side chain that is live on simulated I/Q today —

● Motion gate

Two-pass change detection. Quiet frames skip YOLO. GPU budget goes to movers.

● Detector

YOLO11 / YOLOv8-military / YOLO-World. Classes plus open-vocab prompts (UAV, tank, helicopter).

● Identity

Hyperdimensional item memory on chips. Operator gallery, cosine match — not a 99.8% claim.

● Tracker

EKF + Hungarian association. Optional radar range/bearing update when RF is present.

● RF path

FISTA compressed sensing, Gaussian scale-space, SVD clutter, lightweight peak extraction.

● Runtime

Windows desktop now. ONNX export. TensorRT / INT8 and Jetson are the next SKU, not a redesign.

03 — WHAT IT IS BUILT TO SEE

00

Person

Ground truth from aerial + street sets

01

Bicycle

VisDrone / COCO mapped

02

Car

Including van → car

03

Motorcycle

Ground movers

04

Bus

Large vehicle

05

Truck

Logistics / military truck mapped

06

Airplane

Civil + military airframes

07

Drone

Rotary and fixed-wing UAS

08

Tank

MBT / tracked armor

09

Armored car

APC / AFV family

10

Helicopter

Rotary wing

IN SCOPE

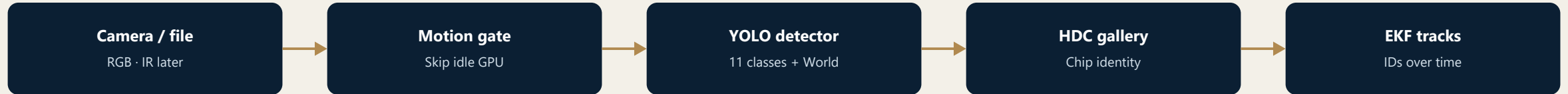
Detect, classify, track, and cue an operator. RGB cameras now. Thermal/IR when labeled frames are supplied. File, webcam, or tower stream. Synthetic coupled radar for software integration today.

OUT OF SCOPE

No fire control, no weapons release, no autonomous engagement, no claimed 99% combat ID. This is a perception layer a company can take into a certified product — not a finished weapon.

Two sensors. One track picture.

OPTICAL PATH



range / bearing update

RF PATH — simulated I/Q now, real recordings later



Software product, not a board spin

CLI already looks like YOLO: predict, track, export, fetch, train.
Integrators can wrap it without learning a new religion.

Coupled scene simulator

Optical movers and synthetic radar returns share the same scene so fusion can be tested before a range day.

Honest RF

Compressed sensing + clutter + cheap topology on peaks. It is a pipeline ready for real I/Q — it is not a fielded radar STAP stack.

A prototype you can run this week. Not a slide-ware stack.

8,755

Labeled train images already merged
(VisDrone + COCO sample)

11

Tactical classes in the training taxonomy

15+

Automated tests (gate, CS, HDC, EKF, JSONL,
video+RF sim)

3060 Ti

CUDA prototype GPU — Jetson is the edge
SKU, not a rewrite

In the repository

- Package: tacyolo · Python 3.10+ · Ultralytics backend
- Weights path: YOLO11, military YOLOv8s, YOLO-World v2
- Training curriculum staged: core aerial → drones → military air/ground → Google Open Images
- Outputs: overlay video, JSONL tracks, ONNX export hook
- Config-driven: YAML for gate, RF, HDC, EKF — no recompile

What that means for a buyer

- Windows desktop first (this lab). Linux / Jetson next.
- No custom PCB. No RF frontend claimed. Software on your camera.
- Public data only so far: VisDrone, COCO, Hugging Face military/drone sets, Open Images planned.
- Private operational video is the fastest accuracy unlock a partner can bring.
- Code is structured for a company team to own, not a one-off Colab.

We would rather lose a meeting than fake a metric.

Defense and security companies have been burned by inflated demos. This page is the product. If a claim is not below, do not write it on a capture slide.

IDENTIFICATION

HDC gallery matching is shipping as cosine-on-chips. There is no 99.8% combat-ID number. Accuracy is gallery- and data-dependent.

RADAR

The RF chain runs on synthetic range-Doppler until the partner provides I/Q or recordings. Adapters are the work, not a science project from zero.

TOPOLOGY

Peak extraction is TDA-lite. It is not a full Vietoris–Rips pipeline and we will not sell it as one.

EDGE PERFORMANCE

INT8 / TensorRT is designed in (export hook) and not yet the production runtime. ONNX is the current portable artifact.

THERMAL / ALL CAMERAS

RGB is trained. Thermal, SWIR, and compressed air-to-ground need labeled frames. Architecture does not block them; data does.

TRAINING DATA

Public sets get you a starting detector. Operational footage from the partner's cameras is what makes a SKU.

A perception layer for companies that already ship the rest.

Counter-UAS & airfields

Detect small UAS and distinguish drones from birds/aircraft in a camera network. Feed an existing alert console.

Perimeter & energy

Towers on pipelines, plants, bases. Motion gate keeps GPU cost sane on mostly-empty scenes.

Defense ISR software

Add a tracker + optional RF cue to an existing Dismount/vehicle workflow without a hardware spin.

Public safety / SAR

People and vehicles from drones or mast cameras. Same model family, different concept of employment.

From lab prototype to a SKU a company can sell.

DAYS 0–90

Pilot

Freeze v0.1 API. Run on partner cameras. Label a private set. TensorRT path on one Jetson. Operator overlay + JSONL into their bus.

DAYS 90–180

Harden

Real I/Q adapter if radar exists. RGB-T if they have dual sensors. Night/weather trial. False-alarm budget. MLOps: retrain without heroics.

DAYS 180–365

Product

Two SKUs (tower + vehicle or UAV payload software). Qualification support. Licensing. Edge performance envelope in writing.

Take this model forward with us.

We are offering the working stack, the training pipeline, and a 90-day joint sprint. You bring cameras, a product owner, and the first customer shape.

WHAT THE COMPANY GETS

- 01 Source software: gated YOLO, HDC matcher, EKF tracker, RF sim chain
- 02 A Windows prototype that already runs on NVIDIA GPU
- 03 A staged training recipe (aerial people/vehicles → drones → armor/air → Open Images)
- 04 Evaluation license for the pilot period, with a path to assignment or exclusive fielding in one vertical
- 05 A written limitation list so capture teams do not over-promise

WHAT WE NEED FROM YOU

- 90-day paid pilot + named product owner
- Operational video (and RF if you have it) under NDA
- One target SKU: tower, vehicle, or payload software
- 1–2 engineers to own integration

NEXT MEETING

90 minutes: live track on your sample video, then a SKU decision. Bring one recording. We bring the model.